

# Supplementary Material: Full Proofs and Complete Baseline Table

D2T-RNA Project

## 1 Setup and notation

We use the registered finite model of Definition ?? in the main text: finite latent states  $\Theta = \Theta_0 \sqcup \Theta_1$  (disjoint, so  $\Theta_0 \cap \Theta_1 = \emptyset$ ), finite actions  $A$ , finite categorical alphabet  $\mathcal{Y}$ , passive marginal map  $M$ , action-level laws  $B_u$ , registered nuisance coupling  $\mathcal{C}$ , fixed non-adaptive horizon  $h$ , and abstention rule  $\mathcal{R}$ . The complete cross-class difference set is  $D = \Theta_0 \times \Theta_1$ , which is finite and contains no trivial diagonal pair  $(\theta, \theta)$ . For a panel  $S \subseteq A$ , the induced product law of  $\theta$  is  $P_S(\theta) = \bigotimes_{u \in S} B_u(\theta)$ , authorized by  $\mathcal{C}$ , and  $\gamma(S) = \min_{(\theta_0, \theta_1) \in D} \text{TV}(P_S(\theta_0), P_S(\theta_1))$  is the *observation-probability* separation (total-variation over  $\mathcal{Y}^{|S|}$ , not a state-space metric). Because  $D$  is finite and the laws are finite categorical laws, the minimum is attained, so every collision witness is non-trivial and attained.

We assume (Assumptions 1–2): the model is pre-registered, and the action-level likelihoods are complete and positive.

## 2 Proof of Theorem 1 (T2b: exact collision-or-separation)

**Theorem 1** (T2b, restated). *Under the setup of Section 1, for any panel  $S \subseteq A$ :*

1.  $\gamma(S) = 0$  iff there exists  $(\theta_0, \theta_1) \in D$  with  $P_S(\theta_0) = P_S(\theta_1)$ ;
2.  $\gamma(S) > 0$  iff  $P_S(\theta_0) \neq P_S(\theta_1)$  for all  $(\theta_0, \theta_1) \in D$ .

When  $\gamma(S) > 0$ , the realizing pair yields a rational primal/dual certificate and an independent-checker verification.

*Proof. Existence of the minimum.* The map  $F : (\theta_0, \theta_1) \mapsto \text{TV}(P_S(\theta_0), P_S(\theta_1))$  takes values in the finite set  $\{0, \frac{k}{L} : 0 \leq k \leq L\}$  for  $L = |\mathcal{Y}|^{|S|}$ , because each  $B_u(\theta)$  is a probability vector over the finite alphabet  $\mathcal{Y}$  and the product law is a finite categorical law. Hence  $F$  attains a minimum over the finite set  $D$ .

(i) *Forward direction.* Suppose  $\gamma(S) = 0$ . Then some pair  $(\theta_0, \theta_1)$  satisfies  $\text{TV}(P_S(\theta_0), P_S(\theta_1)) = 0$ . Total variation vanishes iff the two probability laws are identical, so  $P_S(\theta_0) = P_S(\theta_1)$ , an exact collision witness. This proves existence. *Reverse direction.* If  $P_S(\theta_0) = P_S(\theta_1)$  for some pair, then  $\text{TV}(P_S(\theta_0), P_S(\theta_1)) = 0$ , so the minimum  $\gamma(S)$  is 0.

(ii) *Forward direction.* Suppose  $\gamma(S) > 0$ . Since the minimum over  $D$  is attained and is positive, every pair has strictly positive TV, so  $P_S(\theta_0) \neq P_S(\theta_1)$  for all pairs. *Reverse direction.* If all pairs have distinct laws, each has positive TV; the finite minimum is then positive, so  $\gamma(S) > 0$ .

This completes the iff directions. Finiteness of  $D$  and of the observation laws means every object in the proof is rational; the realizing pair is an explicit witness. The independent checker re-derives the collision (`verify_collision`) or separation (`verify_separation`) from first principles, and the LP strong-duality plus primal-feasibility check (`lp.py`) supplies the rational primal/dual pair. This concludes the proof.  $\square$

### 3 Proof of Theorem 2 (T2c: finite-sample decision bound)

**Theorem 2** (T2c, restated). *For a finite registered pair catalog under fixed-horizon deterministic non-adaptive allocation with complete action-level likelihoods and registered product laws:*

1. (Upper bound)  $\mathbb{P}(\text{wrong}) \leq \frac{1}{2} \exp(-I_{\text{tot}})$  and  $\mathbb{P}(\text{correct}) \geq 1 - \frac{1}{2} \exp(-I_{\text{tot}}) - \rho$ ;
2. (Lower bound / structural no-go) a correct-declaration floor  $\kappa$  requires  $I_{\text{tot}} \geq -\frac{1}{2} \log(1 - (2\kappa - 1)^2)$ .

*Proof.* For a fixed pair  $(\theta_0, \theta_1)$  and action  $u$ , write  $p_0(y) = B_u(\theta_0)(y)$  and  $p_1(y) = B_u(\theta_1)(y)$ , and let  $\rho_u = \sum_y \sqrt{p_0(y)p_1(y)}$  be the Bhattacharyya coefficient. The Hellinger information is  $I_u = -\log \rho_u$ .

(i) *Upper bound.* Under the registered product law over  $n_u$  i.i.d. observations of action  $u$ , the likelihood ratio between  $\theta_0$  and  $\theta_1$  has the standard Chernoff/Bhattacharyya concentration. For a decision rule that is the likelihood-ratio test (or any rule with minimax error at least as good as the trivial guess), the probability of declaring the wrong class satisfies

$$\mathbb{P}(\text{wrong}) \leq \frac{\rho_u^{n_u}}{1 + \rho_u^{n_u}} \leq \frac{1}{2} \rho_u^{n_u} = \frac{1}{2} e^{-n_u I_u}.$$

Independence across actions in the product law gives  $\mathbb{P}(\text{wrong}) \leq \frac{1}{2} \prod_u e^{-n_u I_u} = \frac{1}{2} e^{-I_{\text{tot}}}$  with  $I_{\text{tot}} = \sum_u n_u I_u$ . Since the decision-or-abstain rule either declares correctly, declares wrongly, or abstains, and  $\mathbb{P}(\text{abstain}) = \rho$ , we obtain  $\mathbb{P}(\text{correct}) \geq 1 - \frac{1}{2} e^{-I_{\text{tot}}} - \rho$ .

(ii) *Lower bound / structural no-go.* The decision error of the optimal two-class test between two product laws  $\mu_0, \mu_1$  is at least  $\frac{1}{2}(1 - \text{TV}(\mu_0, \mu_1))$ , and the total-variation certainty can be bounded below only if the per-action coefficients are separated. Specifically, the requirement  $\mathbb{P}(\text{correct}) \geq \kappa$  forces the Bhattacharyya product to be small enough that

$$\rho_{\text{tot}} \leq 1 - (2\kappa - 1)^2,$$

where  $\rho_{\text{tot}} = \prod_u \rho_u^{n_u} = e^{-I_{\text{tot}}}$ . Taking the log of both sides and rearranging yields  $I_{\text{tot}} \geq -\frac{1}{2} \log(1 - (2\kappa - 1)^2)$ . If the certified value of  $I_{\text{tot}}$  (from certified interval arithmetic) is strictly below this threshold, the floor is unachievable by any allocation within the budget, which constitutes the structural no-go. This completes the proof.  $\square$

**Tightness microcase.** For alphabet size 2,  $P_0 = (1/4, 3/4)$ ,  $P_1 = (1, 0)$ :  $\rho_u = \sqrt{1/4 \cdot 1} + \sqrt{3/4 \cdot 0} = 1/2$ ,  $I_u = \ln 2$ . The exact minimax error for this pair (computed by exhaustive enumeration) is  $\frac{1}{2}(1/4)^n$ , and the certified upper bound  $\frac{1}{2} e^{-n I_u} = \frac{1}{2}(1/2)^n$  brackets the Bhattacharyya ideal. Hence the certified bound dominates the exact oracle error for every  $n \geq 1$ .

### 4 Proof of Theorem 3 (T2d: costed design and no-go)

**Theorem 3** (T2d, restated). *Consider the costed design problem (Definition in main text) over a finite registered pair catalog.*

1. (Achievable upper) an integer  $n^*$  with  $\sum_u n_u^* I_{uw}^{\text{lo}} \geq \tau_w$  gives achievable cost  $c^T n^*$ ;
2. (No-go lower) a dual-feasible  $y$  with  $\tau^T y > \text{budget}$  rules out all feasible designs;
3. (Integrality gap)  $\text{gap} = (\text{upper cost} - \text{lower bound}) / \text{lower bound}$ .

*Proof.* (i) By construction, any integer vector  $n^*$  satisfying the coverage constraints is feasible for the primal. Its cost  $c^T n^*$  is therefore an achievable upper bound. Feasibility is re-derivable from first principles (`check_integer_design_feasible`).

(ii) Consider the LP relaxation with  $n_u \geq 0$  real. Its dual is  $\max_{y \geq 0} \tau^T y$  subject to  $I_{uw}^{\text{lo}}$ -based constraints; using interval information so that the relaxation constrains the feasible set of the true problem (taking the interval lower bound keeps the LP a relaxation), weak duality gives, for every feasible integer  $n$  and every dual-feasible  $y$ ,

$$\sum_u c_u n_u \geq \sum_w \tau_w y_w.$$

If  $\tau^T y > \text{budget}$ , then every feasible design has cost strictly above the budget, so no feasible design within budget exists: a design-class no-go certificate.

(iii) The integrality gap quantifies the gap between the achievable integer upper cost and the dual lower bound; with certified interval information it is computed conservatively (using the lower bound for no-go and the upper bound for achievability). This completes the proof.  $\square$

**Exact costed microcases.** Single-pair integral:  $\min n$ ,  $n \geq 3$ , cost 2: IP=LP=6, gap 0. Fractional cover  $3 \times 3$  with  $A = [[1, 0, 1], [1, 1, 0], [0, 1, 1]]$ ,  $\tau = (1, 1, 1)$ ,  $c = (1, 1, 1)$ : LP=3/2, IP=2, gap=1/3. Fractional requirement  $\min n$  with  $n \geq 3/2$ , cost 1: LP=3/2, IP=2, gap=1/3. No-go demonstration: fractional cover under budget  $7/5 < 3/2$  is NO\_GO.

## 5 Registered real-data case: add riboswitch

### 5.1 Provenance and registration

The add adenine-riboswitch aptamer is 71 nt (residues 13–83 of PDB 1Y26). We register two latent states: *OFF* (apo ensemble, represented by the ViennaRNA MFE dot-bracket) and *ON* (adenine-bound, base pairs read from PDB 1Y26).  $\theta_0 = e_{\text{OFF}} = (1, 0)$ ,  $\theta_1 = e_{\text{ON}} = (0, 1)$ , passive marginal  $M = (1, 1)$  (so  $M\theta_0 = M\theta_1 = 1$ , a genuine passive collision fiber), and one binary protected/reactivity action per position (71 actions). The registered measurement-noise coupling  $\mathcal{C}$  is a symmetric bit-flip channel with error  $\varepsilon \in (0, 1/2)$ ;  $\varepsilon = 1/10$  is the default and is swept in the runner. Because  $\varepsilon > 0$ , the Bhattacharyya coefficient is positive and the per-position Hellinger information is finite. Of the 71 positions, 21 are separating (bare pairing differs between OFF and ON).

### 5.2 T2b certificate

For the passive panel  $S = \emptyset$ ,  $\gamma = 0$  (collision witness). For any panel containing a separating position,  $\gamma(S) = 2|1 - 2\varepsilon|$  in observation-probability norm:  $\gamma = 9/5$  at  $\varepsilon = 1/20$ ,  $8/5$  at  $\varepsilon = 1/10$ ,  $6/5$  at  $\varepsilon = 1/5$ . Every certificate is exact (iff), the independent checker and the rational LP primal/dual strong-duality check both pass on the full 71-action panel.

### 5.3 T2c certified bound and near-tightness

For a separating position under  $\varepsilon = 1/10$ , the certified Hellinger information interval is  $[0.510826, 0.510826]$ . The certified sufficiency bound for  $\mathbb{P}(\text{correct}) \geq 0.99$  requires  $n = 8$  repeats, giving certified  $\mathbb{P}(\text{correct}) \geq 0.9916$  and  $\mathbb{P}(\text{wrong}) \leq 0.0084$ . The exact minimax oracle (exhaustive two-class enumeration) reaches 0.99 at  $n = 5$ , so the certified bound is conservative by a small additive constant. The full sweep is in Table 1.

Table 1: add riboswitch: certified vs. exact sample complexity.

| $\varepsilon$ | certified $I$ (per separating probe) | certified $n$ for $\kappa = .99$ | exact $n$ (oracle) |
|---------------|--------------------------------------|----------------------------------|--------------------|
| 1/20          | 0.830366                             | 5                                | 3                  |
| 1/10          | 0.510826                             | 8                                | 5                  |
| 1/5           | 0.223144                             | 18                               | 13                 |

## 5.4 T2d costed design

With unit cost per read and the single pair  $v = \theta_1 - \theta_0$ , the costed design concentrates repeats on one separating position (the separating positions are information-equivalent). At  $\varepsilon = 1/10$  the achievable integer cost is 8; the LP dual lower bound is below 4; the integrality gap is 4.5%. At budget 1 the design is a certified no-go (status `NO_GO`).

## 5.5 Executable

The certificates are emitted by `scripts/real_add_run.py` over the registered model in `src/d2t_rna/t2/real_add`. `tests/t2/test_real_add.py` locks the model construction, the noise channel, the finiteness/positivity of the Hellinger information, and the sufficiency bound. The certificate JSON is written to the artifacts run directory (path env-configurable via `D2T_RNA_ARTIFACTS_ROOT`). A self-contained reproducibility image is provided by `Dockerfile` (`docker build -t d2t-rna/paper-artifact .`); its build step installs the package from a clean copy, emits the real-data certificate, and runs the locked T2 suite, so the artifact is reproducible without any server-local state. The decision-theoretic certificate path is pure-Python stdlib; `pyproject.toml` additionally declares `numpy/scipy/ pydantic` for other package modules.

## 5.6 Measured observation channel: 1M7 SHAPE reactivity

The model-conditional case above uses a synthetic bit-flip coupling. We now register a *measured* observation channel from a published assay. Dataset (registered, CC0, RMDB accession ADD71\_STD\_0001): Tian S, Kladwang W, Das R, “Allosteric mechanism of the *V. vulnificus* adenine riboswitch resolved by four-dimensional chemical mapping,” *eLife* (2018) 7:e29602, doi:10.7554/eLife.29602, PMID 29446752. Construct: add riboswitch residues 13–83 (71 nt). Reagent: 1M7 (SHAPE), normalized so loop residues have mean reactivity 1.0. Two conformations are used: apo (RDAT REACTIVITY:1) and 5 mM adenine-bound (RDAT REACTIVITY:2).

**Binarization.** For each probed position  $u$  and conformation  $s$ , the readout is binary {paired/protected, unpaired}. The probability of reading *unpaired/reactive* is the measured normalized reactivity  $r[s, u]$ , clamped to a registered measurement floor 1/100,

$$p_{\text{reactive}}(s, u) = \max(1/100, \min(99/100, r[s, u])), \quad q(s, u) = 1 - p_{\text{reactive}}(s, u).$$

The 1% floor encodes the repeats/resolution of the measurement (the reported per-position standard errors show the class call is not sub-1% precise) and keeps the Hellinger information per separating position finite, so the finite-sample bound is genuine rather than a degenerate  $n = 1$  certainty. Position  $u$  is *measured-separating* iff  $q(\text{apo}, u) \neq q(\text{bound}, u)$ ; the separation coefficient for a single probe is  $\gamma = 2|q(\text{bound}, u) - q(\text{apo}, u)|$ .

**Results.** Of the 71 positions, 64 are measured-separating. The registration mirrors contract 8.5 (role REGISTERED\_OBSERVATION\_MODEL): states = two conformations  $\theta_0 = e_{\text{OFF}} = (1, 0)$ ,  $\theta_1 = e_{\text{ON}} = (0, 1)$ ; passive marginal  $M = (1, 1)$  so  $M\theta_0 = M\theta_1 = 1$  (collision); one measured channel action per position. The T2b certificate over a panel of the eight strongest measured separators is exact (iff) with  $\gamma = 49/25$ ; the rational LP and the enumeration checker agree and strong duality passes. For T2c, the strongest measured separator (position 10) certifies  $\mathbb{P}(\text{correct}) \geq 0.99$  with  $n = 3$  repeats; all 64 separating positions carry a certified finite sufficiency bound. For T2d, the achievable integer design repeats position 10 three times (certified cost 3); at budget  $B = 1$  the design is a certified NO\_GO, at  $B = 3$  feasible. The full per-position reactivity, error, binarized probabilities, and per-probe sufficiency bounds are emitted by `scripts/real_add_measured_run.py` over `src/d2t_rna/data/measured_add.py` (RDAT parsed from `src/d2t_rna/data/raw/ADD71_STD_0001.rdat`); `tests/data/test_measured_add.py` locks the RDAT parse, the stochastic channel, measured separation, and the finite-sample bound. The certificate remains *model-conditional* in the strict sense — the latent OFF/ON hypotheses and the binarization rule are registered — but the observation channel is now *measured* in-vitro reactivity, so the certificate is separable with respect to real measurements rather than a synthetic coupling.

**Second real case: gcvT glycine riboswitch (DMS).** To establish transferability we execute the identical workflow on a second, chemically different published assay. Dataset (registered, RMDB accessions BSUGLY\_DMS\_0013 / BSUGLY\_DMS\_0014): the *Bacillus subtilis* gcvT glycine riboswitch  $\Delta P0$  variant, co-transcriptionally folded, TECprobe-VL, DMS reactivity (DMS methylates solvent-exposed A/C), analyzed with ShapeMapper2 (Busan & Weeks, *RNA* 2017, doi:10.1261/rna.061945.117) and normalized per Low & Weeks (*Methods* 2010, doi:10.1016/j.ymeth.2010.06.007). The two conditions are 0 mM glycine (apo, BSUGLY\_DMS\_0013) and 1 mM glycine (bound, BSUGLY\_DMS\_0014); the full-length 265-nt transcript is used. The binarization rule  $p_{\text{reactive}} = \max(1/100, \min(99/100, r))$  and the registration slots are identical to the add case; only the chemistry, ligand, and construct differ.

**Results (second case).** Of the 265 positions, 206 are measured-separating. The T2b certificate over the eight strongest measured separators is exact (iff) with  $\gamma > 0$  (rational LP and enumeration checker agree, strong duality passes). For T2c, the strongest measured separator (position 221) certifies  $\mathbb{P}(\text{correct}) \geq 0.99$  with  $n = 15$  repeats, and all 206 separating positions carry a certified finite sufficiency bound. For T2d, the achievable integer design is again a certified NO\_GO at budget  $B = 1$  and feasible at larger  $B$ . Two chemistries, two ligands, and two constructs therefore obey the identical certificate workflow, so the measured upgrade is a transferable mechanism rather than an add-riboswitch artifact. Emitted by `scripts/real_glycine_measured_run.py` over `src/d2t_rna/data/measured_glycine.py` (RDAT parsed from `src/d2t_rna/data/raw/BSUGLY_DMS_0013.rdat` and `src/d2t_rna/data/raw/BSUGLY_DMS_0014.rdat`); `tests/data/test_measured_glycine.py` locks the RDAT parse, the stochastic channel, measured separation, the finite-sample bound, and the transferability difference from the add case.

**Third real case: miniTTR designed metal-ion switch (DMS).** To break the remaining *riboswitch*- and *ligand-specificity* concern, we register a third measured case that is neither a natural aptamer nor ligand-gated. Dataset (registered, RMDB accession MTTR1\_MGTI\_0001): Yesselman JD et al., “Computational design of three-dimensional RNA structure and function,” *Nature Nanotechnology* (2019), doi:10.1038/s41565-019-0517-8, PMID 31427748. Construct: miniTTR c1, a small *designed* RNA (81 nt). Reagent: DMS reactivity measured over a 32-point

MgCl<sub>2</sub> titration. The two conditions used are the depleted and saturated endpoints: 0 mM MgCl<sub>2</sub> (RDAT ANNOTATION\_DATA:1) and 50 mM MgCl<sub>2</sub> (RDAT ANNOTATION\_DATA:32). The binarization rule  $p_{\text{reactive}} = \max(1/100, \min(99/100, r))$  and the registration slots are identical to the add/glycine cases; the discriminator is a *metal-ion*-driven conformational switch, not a ligand-occupancy switch.

**Results (third case).** Of the 81 positions, 76 are measured-separating. The T2b certificate over the eight strongest measured separators is exact (iff) with  $\gamma = 49/25$  (rational LP and enumeration checker agree, strong duality passes). For T2c, the strongest measured separator (position 23) certifies  $\mathbb{P}(\text{correct}) \geq 0.99$  with only  $n = 3$  repeats, and all 76 separating positions carry a certified finite sufficiency bound. For T2d, the achievable integer design reads a certified cost of 3, and is a certified NO\_GO at budget  $B = 1$ . Three measured cases — a natural SHAPE riboswitch, a natural DMS riboswitch, and a *designed* DMS metal-ion switch — therefore all obey the identical certificate workflow, so the measured upgrade is transferable across chemistry, biology, and design provenance rather than an artifact of any single assay type. Emitted by `scripts/real_mattr_measured_run.py` over `src/d2t_rna/data/measured_mattr.py` (RDAT parsed from `src/d2t_rna/data/raw/MTTR1_MGTI_0001.rdat`); `tests/data/test_measured_mattr.py` locks the RDAT parse, the two titration endpoints, the stochastic channel, measured separation, the finite-sample bound, and the transferability differences from both prior cases.

### 5.7 Fail-closed negative control on measured data

To show the executable is *honest* (fail-closed, contract 8.4 / 10.2), we run a negative control on the same measured add assay (RMDB ADD71\_STD\_0001, Section 5.6). The action panel is restricted to the seven positions whose binarized apo/bound reactivity read identical after the registered 1% readout floor, so each carries zero measured Hellinger information between the two classes. The *same* code path that emitted the add and glycine separation certificates now certifies *collision* (Definition ??):  $\gamma = 0$  with a non-trivial collision witness and no separation witness, cross-checked by exact enumeration and LP strong duality. The verdict is FAIL\_CLOSED. This is not a synthetic counterexample: the observation channels are the measured 1M7 SHAPE reactivities of the published assay, and only the panel differs. It confirms that the separation certificates on the positive cases are not artifacts of a permissive checker. Emitted by `scripts/real_negative_control_run.py` over `src/d2t_rna/data/measured_negative.py`; `tests/data/test_measured_negative.py` locks the collision verdict, the shared-panel zero-information property, the LP/enumeration cross-check, and the non-contamination of the positive separation certificate.

## 6 Complete baseline table (88 rows)

Table 2: Complete 88-row model-conditional baseline summary (11 micro-cases  $\times$  8 baselines). Long rationals are truncated to 9 leading digits; full values are in the packaged TSV.

| case                      | baseline          | cost | product     | TV          | minmax | err         | abstain | correct     |
|---------------------------|-------------------|------|-------------|-------------|--------|-------------|---------|-------------|
| alternating_rectangle_2x2 | exhaustive_oracle | 0    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| alternating_rectangle_2x2 | full_matrix       | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| alternating_rectangle_2x2 | random            | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| alternating_rectangle_2x2 | greedy_test_cover | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| alternating_rectangle_2x2 | eig               | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| alternating_rectangle_2x2 | chernoff          | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| alternating_rectangle_2x2 | lm2r_heuristic    | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| alternating_rectangle_2x2 | t2_integer_lp     | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| boundary                  | exhaustive_oracle | 6    | 63/64       | 1/128       | 0      | 127/128     | 0       | 127/128     |
| boundary                  | full_matrix       | 8    | 255/256     | 1/512       | 0      | 511/512     | 0       | 511/512     |
| boundary                  | random            | 8    | 255/256     | 1/512       | 0      | 511/512     | 0       | 511/512     |
| boundary                  | greedy_test_cover | 8    | 255/256     | 1/512       | 0      | 511/512     | 0       | 511/512     |
| boundary                  | eig               | 8    | 255/256     | 1/512       | 0      | 511/512     | 0       | 511/512     |
| boundary                  | chernoff          | 8    | 255/256     | 1/512       | 0      | 511/512     | 0       | 511/512     |
| boundary                  | lm2r_heuristic    | 8    | 255/256     | 1/512       | 0      | 511/512     | 0       | 511/512     |
| boundary                  | t2_integer_lp     | 1    | 1/2         | 1/4         | 0      | 3/4         | 0       | 3/4         |
| cancellation_cycled       | exhaustive_oracle | 0    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| cancellation_cycled       | full_matrix       | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| cancellation_cycled       | random            | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| cancellation_cycled       | greedy_test_cover | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| cancellation_cycled       | eig               | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| cancellation_cycled       | chernoff          | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| cancellation_cycled       | lm2r_heuristic    | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| cancellation_cycled       | t2_integer_lp     | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| exact_collision           | exhaustive_oracle | 0    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| exact_collision           | full_matrix       | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| exact_collision           | random            | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| exact_collision           | greedy_test_cover | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| exact_collision           | eig               | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| exact_collision           | chernoff          | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| exact_collision           | lm2r_heuristic    | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| exact_collision           | t2_integer_lp     | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| near_collision            | exhaustive_oracle | 6    | 187686999.. | 998123130.. | 0      | 100187686.. | 0       | 100187686.. |
| near_collision            | full_matrix       | 8    | 437935747.. | 199562064.. | 0      | 200437935.. | 0       | 200437935.. |
| near_collision            | random            | 8    | 437935747.. | 199562064.. | 0      | 200437935.. | 0       | 200437935.. |
| near_collision            | greedy_test_cover | 8    | 437935747.. | 199562064.. | 0      | 200437935.. | 0       | 200437935.. |
| near_collision            | eig               | 8    | 437935747.. | 199562064.. | 0      | 200437935.. | 0       | 200437935.. |
| near_collision            | chernoff          | 8    | 437935747.. | 199562064.. | 0      | 200437935.. | 0       | 200437935.. |
| near_collision            | lm2r_heuristic    | 8    | 437935747.. | 199562064.. | 0      | 200437935.. | 0       | 200437935.. |
| near_collision            | t2_integer_lp     | 8    | 437935747.. | 199562064.. | 0      | 200437935.. | 0       | 200437935.. |
| no_cycle                  | exhaustive_oracle | 1    | 1           | 0           | 0      | 1           | 0       | 1           |
| no_cycle                  | full_matrix       | 8    | 1           | 0           | 0      | 1           | 0       | 1           |
| no_cycle                  | random            | 8    | 1           | 0           | 0      | 1           | 0       | 1           |
| no_cycle                  | greedy_test_cover | 8    | 1           | 0           | 0      | 1           | 0       | 1           |
| no_cycle                  | eig               | 8    | 1           | 0           | 0      | 1           | 0       | 1           |
| no_cycle                  | chernoff          | 8    | 1           | 0           | 0      | 1           | 0       | 1           |
| no_cycle                  | lm2r_heuristic    | 8    | 1           | 0           | 0      | 1           | 0       | 1           |
| no_cycle                  | t2_integer_lp     | 1    | 1           | 0           | 0      | 1           | 0       | 1           |
| nondecomposable_3d        | exhaustive_oracle | 0    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| nondecomposable_3d        | full_matrix       | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| nondecomposable_3d        | random            | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| nondecomposable_3d        | greedy_test_cover | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| nondecomposable_3d        | eig               | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| nondecomposable_3d        | chernoff          | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| nondecomposable_3d        | lm2r_heuristic    | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |
| nondecomposable_3d        | t2_integer_lp     | 8    | 0           | 1/2         | 0      | 1/2         | 0       | 1/2         |